PPIGM x LPDP Manchester Seminar

`Ggplot2`: How to (Elegantly) Visualise Your Research

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The University of Manchester

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Outline

- Why Visualise Data?
- What is a good data visualisation?
- Understanding `ggplot2`
  - Data frame basics
  - Processing the data
  - `ggplot` layers
- Hands-on
- Q&A

- Understanding effective visual principle
- Understand how to visualise data
- Can implement ggplot2 library in their daily task (especially dissertation)

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Data Visualisation

Why Visualise Data?

Month	Received	Processed
Jan	160	160
Feb	184	184
Mar	241	237
Apr	149	148
May	180	181
Jun	161	132
Jul	150	123
Aug	202	156
Sep	160	126
Oct	139	104
Nov	149	124
Dec	177	140

How do we analyse this data?

Year	Month	Received	Processed
2024	Jan	184	184
2024	Feb	146	187
2024	Mar	162	188
2024	Apr	171	175
2024	May	174	173
2024	Jun	192	154
2024	Jul	167	139
2024	Aug	152	198
2024	Sep	201	153
2024	Oct	153	174
2024	Nov	181	174
2024	Dec	177	156
2025	Jan	160	160
2025	Feb	184	184
2025	Mar	241	237
2025	Apr	149	148
2025	May	180	181
2025	Jun	161	132
2025	Jul	150	123
2025	Aug	202	156
2025	Sep	160	126
2025	Oct	139	104
2025	Nov	149	124
2025	Dec	177	140

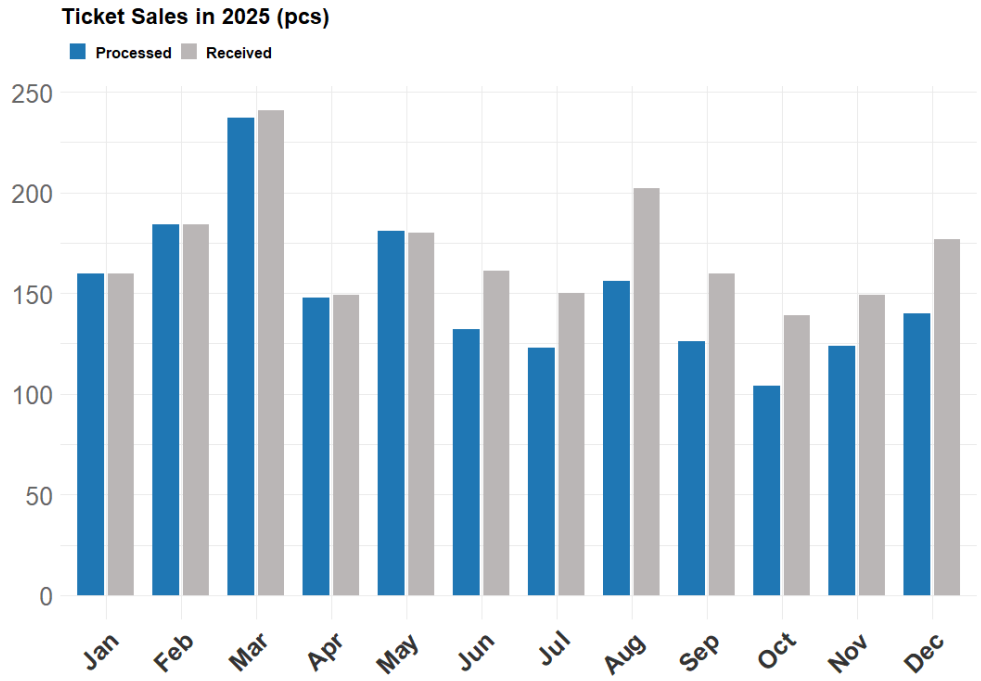
??????

☹ ☹ ☹ ☹ ☹ ☹

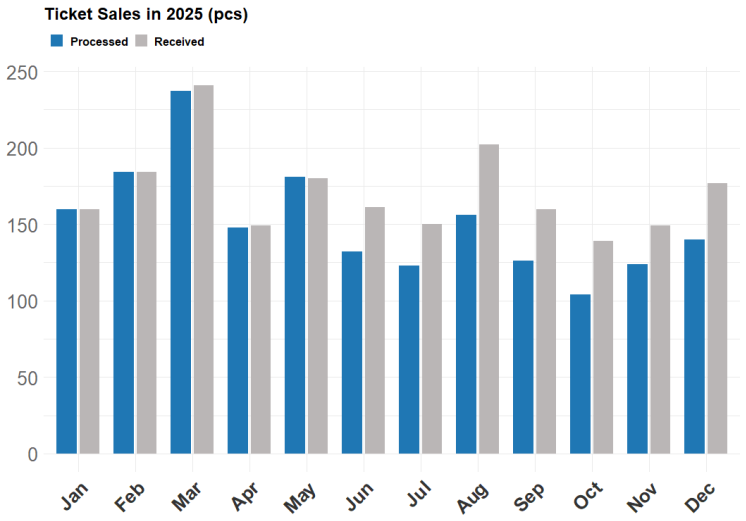
This is where visualisation is important

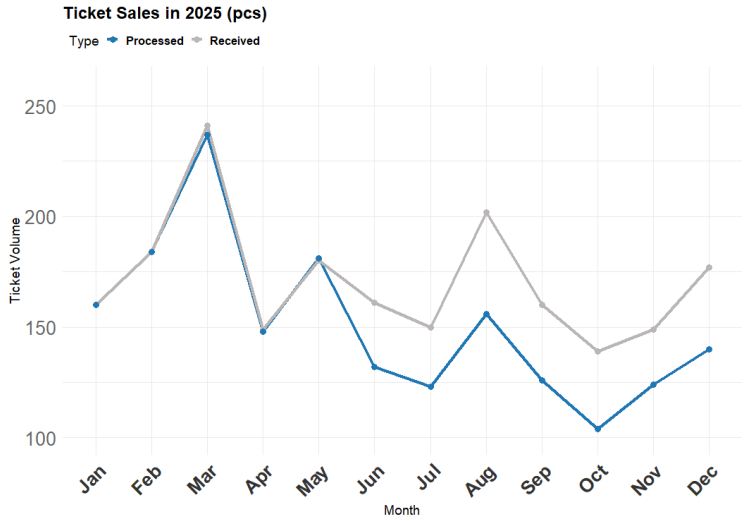
Why Visualise Data?

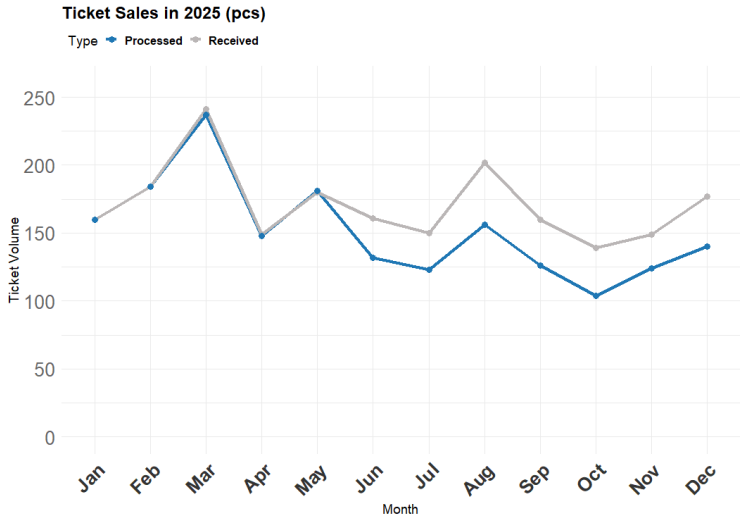
- To make comparisons **easier**
- To reveal patterns and **trends**
- To communicate insights **clearly**
- To direct attention to what **matters**



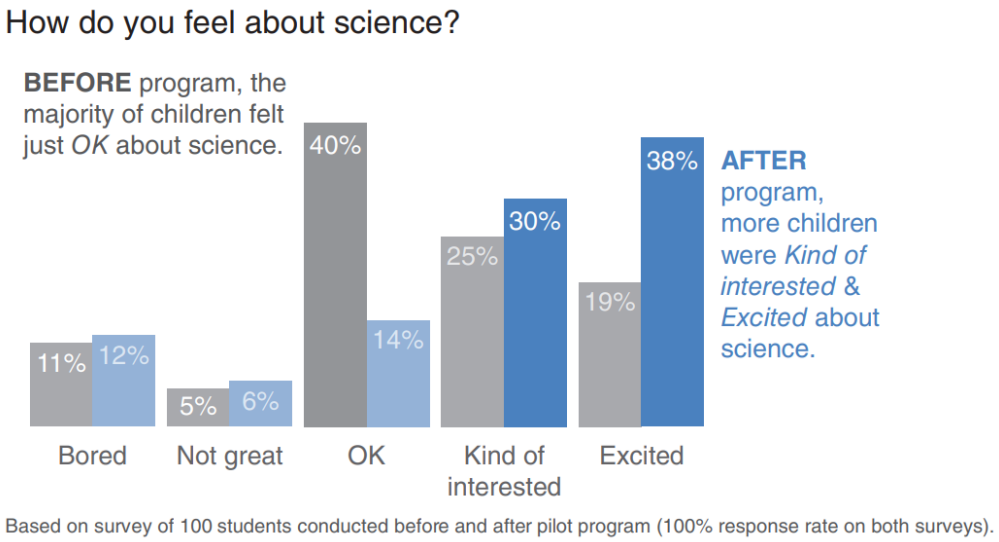
What do you think about this graph?



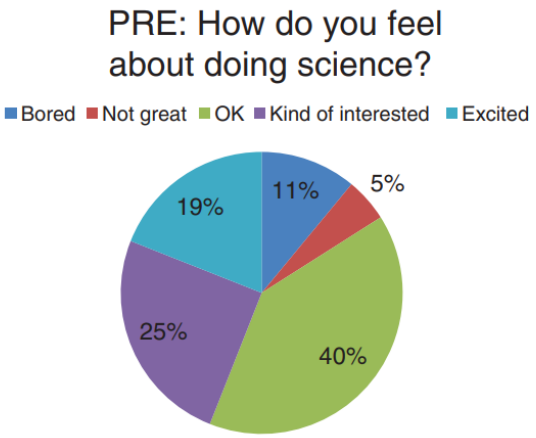


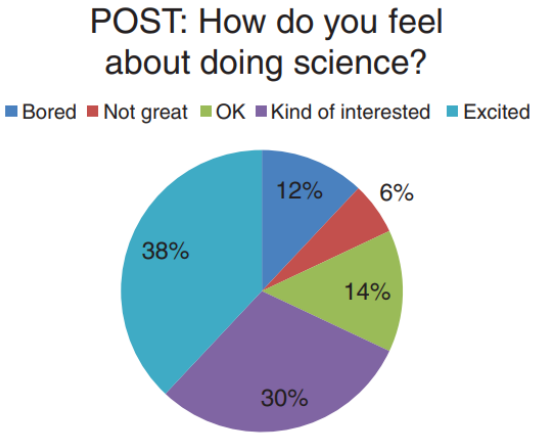


Which One is Better?



Based on survey of 100 students conducted before and after pilot program (100% response rate on both surveys).





What makes a good chart?

- It has a **clear** purpose
- It uses the **right** chart type
- It removes unnecessary **clutter**
- It **directs attention** to the main insight

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Principles of Data Visualisation

- Understand the context
- Choose an appropriate visual display
- Eliminate clutter
- Focus attention where you want it
- Think like a designer
- Tell a story

Understand the context

- Who is your audience?
  - Student? Business partner? High level? Community?
  - Ensure they hear your message
- What do you need them to know or do?
  - How you want your audience to act
  - How you will communicate to them and the overall tone that you want to set for your communication
- How can you use data to help make your point?
  - Exploratory: understand the data and figure out what might be noteworthy or interesting to highlight to others.
  - Explanatory: communicating analysis to audience. have a specific thing you want to explain, a specific story you want to tell

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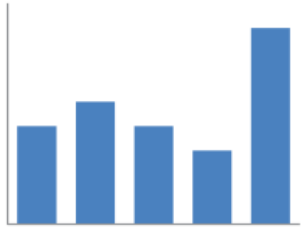
Choose an appropriate visual display

91%

Simple text

Scatterplot



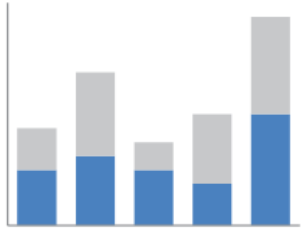
Vertical bar

Horizontal bar

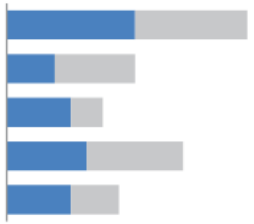
	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

Table

Line



Stacked vertical bar

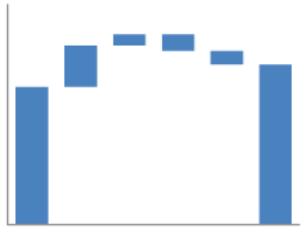


Stacked horizontal bar

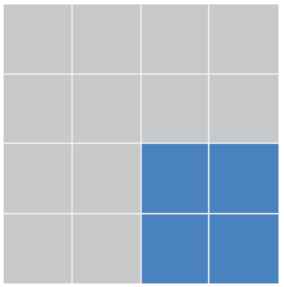
	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

Heatmap

Slopegraph



Waterfall



Square area

Declutter and Attention Focus

- Declutter
  - Remove unnecessary visual elements
  - Reduce redundant labels, legends, and gridlines
  - Avoid excessive colours and decoration
  - Keep the figure clean and readable
- Attention Focus
  - Guide the reader to the main result
  - Use visual hierarchy, contrast, and ordering carefully
  - Add clear labels and brief annotations where needed
  - Make the key takeaway easy to identify

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Visualise your results using `ggplot2`

What is ggplot2

- A package in **R** for building data visualisations.
- Based on the **Grammar of Graphics**.
- Turns data into charts through clear layers.
- Map variables to visual elements like **x**, **y**, **colour**, and **size**.
- Makes plots easier to customise, improve, and reuse.

What kind of data does ggplot2 need?

columns = variables

rows = observations

id	sex	education	income
1	M	Senior high school	30,000
2	F	PhD	48,000
3	F	Under-graduate	50,000
4	M	Under-graduate	55,000
Categorical	Categorical	Categorical	Numerical

- Numerical: `real`, `integer`
- Categorical: `char`, `str`, `factor`
- Raw file format: everything (`csv`, `stata`, `xls`, etc.)
- Remember: no merged column!

ggplot2: Grammar of Graphics Plot

`Data`

Data that you have

`aes()`

What variables to use and mapped to where.

`x, y, group` (incl. `fill, colour`), etc.

`geom_*`

What type of graphic to use

Bar plot (`geom_bar, geom_col`)

Scatter (`geom_point`)

Box plot (`geom_boxplot`)

Line (`geom_line`), etc.

Simple example:

```
ggplot(data = df,
       aes(x = ...,
           y = ...)) +
       geom_...()
```

optional: `facet, label, theme_*`

Needed for "touch-up" the plot

The Optionals

- Needed for “touch-up” the plot
- `facet`
  - Split plot into categories
- `label`
  - Naming the main names (axes and title)
- `scale_*`
  - Manage more detailed names (axes ticks, legend ticks, etc.)
- `theme_*`
  - Manage non-data elements (e.g. font, background, grid, legend position)

Hands-on

- Open https://github.com/rulnasution/seminars/
- Go to ppixlpdp-ggplot folder
- Download the hands-on.R

Closing Statement

- Think first
  - What message you want to deliver
- Choose plot
  - Comparison: Bar plot
  - Distribution: Histogram/Boxplot
  - Relationship: Scatter plot
  - Time-series: Line plot
  - etc.
- And ggplot2 helps us turn both into something clear and elegant.
- More ggplot galleries (script provided)
  - https://r-graph-gallery.com/

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Thank you

Terima kasih

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